Uncertainty-Aware NeRF-SLAM

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***Abstract*—Neural Radiance Field (NeRF) has become a powerful and trending representation for 3D environments. Recent works [1], [2] leverage its capability in Simultaneous Localization and Mapping (SLAM) to reconstruct environments and estimate camera poses online. However, online SLAM must be both fast and accurate given a fixed number of samples, making efficient use of limited information critical for final performance. In this project, we incorporate uncertainty with covisibility in the keyframe selection process for NeRF optimization, which results in improved localization accuracy compared to the original implementation. Furthermore, we vectorize the uncertainty calculation to make it applicable in real time. Future work could explore more efficient uncertainty measures as a direction for further improvement. Code is available at https://github.com/tzuchieh-robotics/Uncertainty-Aware-NeRF-SLAM and a demonstration can be found at https://stunning-syrniki-05e831.netlify.app/.**

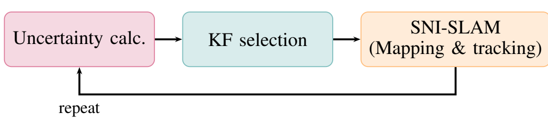

Fig. 1: Overview of our uncertainty-aware NeRF-SLAM pipeline.

I. Introduction

NERF-SLAM is a relatively novel paradigm that extends Neural Radiance Fields (NeRF) to simultaneous localization and mapping. The core idea is to train a neural network online to jointly reconstruct the scene geometry and estimate camera trajectories in real time. As a result, it requires substantial computational resources, efficient algorithm design, and effective utilization of limited data.

Given the limited number of available samples during online operation, it is crucial to maximize the effectiveness of each NeRF update while maintaining real-time performance. In practice, the NeRF-SLAM pipeline often operates in an underfitting regime due to time and data constraints. Therefore, the strategy used to select training samples can significantly affect convergence speed and, consequently, the overall SLAM performance.

The original implementation in [1] adopts covisibility as the criterion for keyframe selection. Frames with high covisibility provide strong spatial constraints and are therefore prioritized for updating the neural field. However, from our experiments, we observe that the uncertainty of individual frames gradually decreases throughout training. This motivates us to incorporate uncertainty into the keyframe selection process.

Specifically, we use entropy as a measure of frame-wise uncertainty and prioritize frames with higher uncertainty for network updates. This encourages the model to focus on less certain regions, leading to more informative updates.

Introducing uncertainty, however, incurs additional computational overhead, particularly in entropy calculation and sample selection. This overhead is undesirable for real-time SLAM systems. To address this, we vectorize the entire process by converting frame-wise operations into batch computations. As a result, our implementation achieves significantly improved efficiency and remains only marginally slower than [1].

Overall, our contributions are as follows: We propose an uncertainty-aware keyframe selection strategy for NeRF-SLAM that integrates entropy-based uncertainty with covisibility, enabling more informative and targeted network updates. We show that prioritizing high-uncertainty frames improves convergence behavior under the inherently underfitting regime of online NeRF-SLAM. We design a fully vectorized implementation for entropy computation and sample selection, significantly reducing computational overhead and preserving near real-time performance. We empirically validate that our method improves reconstruction quality and SLAM accuracy compared to covisibility-only baselines.

We also found that selecting frames with higher uncertainty results higher localization accuracy, our guess is these frames are more informative for updating camera position.

II. Related Work

Neural Radiance Fields (NeRF) [3] introduced a powerful framework for novel view synthesis by representing scenes as continuous volumetric functions parameterized by multilayer perceptrons. Subsequent works improved training speed [4], scene editing, and generalization. Implicit surface representations based on Signed Distance Fields (SDF), such as NeuS [5], further extended this paradigm to geometry reconstruction by coupling volume rendering with neural implicit surfaces, enabling higher-quality mesh extraction. These advances laid the foundation for applying NeRF-style representations to online mapping problems.

A growing line of work has explored integrating neural representations into SLAM pipelines. iMAP [6] was among the first to use a single MLP as the scene representation for dense RGB-D SLAM, jointly optimizing geometry and camera poses online. NICE-SLAM [7] addressed scalability by replacing the global MLP with a hierarchical feature grid, enabling reconstruction of larger scenes. Subsequent systems such as ESLAM [8] and Co-SLAM [9] further improved efficiency by adopting hybrid representations combining explicit feature grids with implicit decoders. SNI-SLAM [1] extended this direction by incorporating semantic understanding via DINOv2-based features and triplane scene representations, enabling simultaneous dense geometry, appearance, and semantic reconstruction. [10] used visual-inertial SLAM with NeRF in real time. They treated changes before and after loop closure as dynamic scene transformations, allowing previous optimization information to be reused, which resulted in better camera poses and a more consistent map overall.

Alongside these reconstruction advances, uncertainty estimation has been studied in the context of neural rendering to identify informative or under-reconstructed regions. ActiveNeRF [11] proposed using Shannon entropy of volume rendering weights as a proxy for geometric uncertainty, and used it to guide active view selection for more efficient scene learning. CF-NeRF [12] incorporated Bayesian uncertainty into radiance field training to quantify prediction confidence. This connects naturally to the problem of keyframe selection in SLAM, where determining which frames are retained and used for map updates is critical to overall performance. Classical visual SLAM systems such as ORB-SLAM [13] select keyframes based on feature overlap and tracking quality, while neural SLAM systems commonly rely on covisibility as the selection criterion [1], [7]. However, covisibility alone does not account for the model's uncertainty about the space, which can make the model keep optimizing regions with high covisibility but ignoring regions with high uncertainty that contains lots of information for the model. Our method augments covisibility with rendering uncertainty to preferentially select frames that are most informative given the current neural field, addressing this gap.

III. Reproduction

SNI-SLAM [1] is a neural implicit SLAM system that jointly performs dense mapping and camera tracking using triplane-based scene representation. The system maintains three sets of feature planes for geometry, color, and semantics respectively, which are queried via trilinear interpolation to decode SDF values, RGB colors, and semantic labels at arbitrary 3D locations. A DINOv2-based segmentation network provides per-frame semantic features that are fused into the scene representation during mapping.

The system runs two parallel threads: a **Mapper** that optimizes the scene representation by minimizing SDF, color, depth, feature, and semantic losses over selected keyframes, and a **Tracker** that estimates camera poses by minimizing rendering loss with the scene representation frozen. Reproduction was straightforward. We cloned the official repository, downloaded the pretrained DINOv2 weights, and ran the system on the Replica dataset with minimal setup. No modifications to the training procedure or hyperparameters were required. The system ran successfully out of the box, producing dense semantic reconstructions and camera trajectory estimates consistent with the results reported in the original paper. The reconstructed mesh and rendering result are shown in Fig. **??** and Fig. 5.

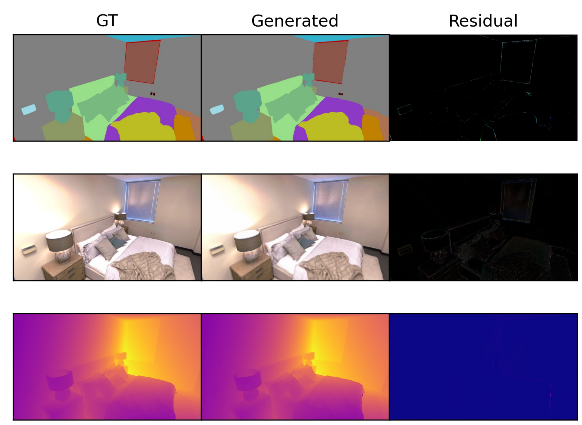

Fig. 2: Rendered frame

Fig. 3: Reproduction results on Replica room_1. Left: rendered RGB frame. Right: reconstructed semantic mesh.

IV. Methodology

In this section, we present our uncertainty-aware keyframe selection framework for NeRF-SLAM. Our method extends the standard pipeline by incorporating entropy-based uncertainty alongside covisibility to guide the selection of informative frames for updating the neural field.

A. Uncertainty Calculation

To quantify the uncertainty of each frame for keyframe selection, we adopt the formulation in [11], where Shannon entropy is used to measure uncertainty along each ray. Specifically, the ray-wise weights are derived from

Fig. 4: Rendered frame

Fig. 5: Reproduction results on Replica room_1. Left: rendered RGB frame. Right: reconstructed semantic mesh.

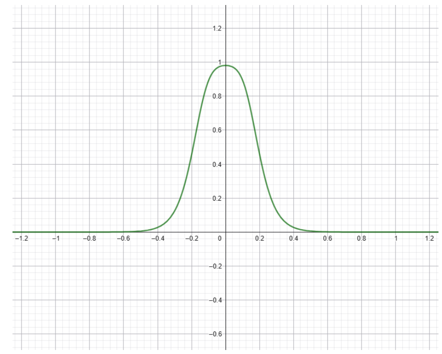

Fig. 6: Visualization of opacity α derived from SDF values. A sharper transition indicates more confident surface prediction.

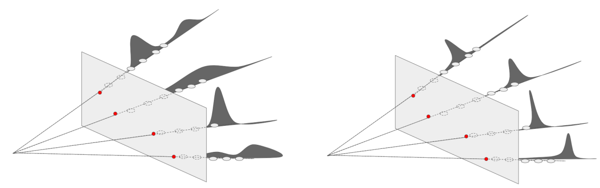

Fig. 7: Entropy visualization. Rays are sampled across the image, and entropy is computed from the weight distribution along each ray. The right example shows higher uncertainty (higher entropy) compared to the left.

the Signed Distance Field (SDF), which is jointly predicted with semantics and other features. Following [1], the opacity α_i is computed from the SDF value as:

$$\alpha_i = 1 - \exp\left(-\beta \cdot \sigma(-\mathrm{SDF}_i \cdot \beta)\right) \tag{1}$$

where $\sigma(\cdot)$ denotes the sigmoid function and β is a learnable parameter controlling the sharpness of the surface transition. As illustrated in Fig. 6, higher opacity indicates a higher probability of a point lying on the surface. For a well-trained SDF, the opacity distribution is expected to be sharply concentrated around the true surface. In contrast, uncertain regions exhibit a more spread-out distribution of weights along the ray.

Based on these weights, we compute the Shannon entropy for each ray as:

$$H = -\sum_i w_i \log(w_i + \epsilon) \tag{2}$$

where w_i represents the normalized contribution of each sampled point along the ray. To estimate frame-level uncertainty, we randomly sample 64 pixels per frame and perform ray casting. The final uncertainty of a frame is defined as the mean entropy across all sampled rays. As shown in Fig. 7, rays with more distributed weights tend to produce higher entropy values.

B. *Vectorization*

To efficiently compute uncertainty across all keyframes, we vectorize the uncertainty estimation process. Instead of sequentially computing uncertainty for each keyframe, we batch all keyframe rays into a single forward pass through the scene representation.

Specifically, given N_{kf} keyframes, we sample N_r rays per keyframe by randomly sampling pixel locations. All rays are collected into a single batch of size $N_{kf} \times N_r$, and depth-guided sampling is applied uniformly to generate 3D query points along each ray following the same strategy as the mapping process:

$$z_{\text{free}} = \{t \cdot 1.2 \cdot d_{gt}\}_{t\in[0,1]}, \quad z_{\text{surf}} = \{d_{gt} + t \cdot 3\tau - 1.5\tau\}_{t\in[0,1]} \tag{3}$$

where d_{gt} is the ground truth depth and τ is the truncation distance. The batch of points is then passed through the SDF decoder f_θ in a single forward pass:

$$\{\mathrm{SDF}_i\} = f_\theta(\{p_i\}), \quad i = 1, \ldots, N_{kf} \times N_r \times N_s \tag{4}$$

where N_s is the number of samples per ray. The resulting SDF values are converted to volume rendering weights w, from which the per-keyframe uncertainty is computed as:

$$U_k = \frac{1}{N_r} \sum_{r=1}^{N_r} H(w_r), \quad H(w) = -\sum_i w_i \log(w_i + \epsilon) \tag{5}$$

This vectorized formulation reduces the number of decoder forward passes from N_{kf} to 1, yielding a significant speedup as the number of keyframes grows.

TABLE I: Runtime comparison.

Method	FPS	Relative to Baseline
Baseline (SNI-SLAM)	0.88	100%
+ Uncertainty Estimation	0.46	52%
+ Vectorization	0.68	77%

C. Combined with Covisibility

To prioritize keyframes that are both geometrically relevant and informationally uncertain, we combine covisibility with multi-modal uncertainty into a unified selection score. For each candidate keyframe i, we define:

$$s_i = c_i \cdot \left(w_{\text{geo}}\, \hat{u}_i^{\text{geo}} + w_{\text{depth}}\, \hat{u}_i^{\text{depth}} + w_{\text{sem}}\, \hat{u}_i^{\text{sem}} \right) \quad (6)$$

where $c_i \in [0, 1]$ is the covisibility score (fraction of projected points visible in keyframe i), and $\hat{u}_i^{\text{geo}}$, $\hat{u}_i^{\text{depth}}$, $\hat{u}_i^{\text{sem}}$ are min-max normalized uncertainties for geometry, depth, and semantics, respectively.

Geometric uncertainty is measured by the entropy of volume rendering weights:

$$u_i^{\text{geo}} = -\sum_k w_{ik} \log(w_{ik} + \epsilon) \quad (7)$$

Depth uncertainty is the mean absolute error between rendered and observed depth, and semantic uncertainty is the entropy of the predicted class distribution:

$$u_i^{\text{sem}} = -\sum_c p_{ic} \log(p_{ic} + \epsilon) \quad (8)$$

Keyframes are then sampled from the softmax distribution over s_i, ensuring that frames with high overlap *and* high uncertainty are preferentially selected for joint optimization.

V. Experiments and Results

We evaluate our method on the Replica dataset [14] room_1 sequence and compare against the SNI-SLAM baseline [1]. All experiments are conducted on an NVIDIA Tesla V100 GPU. For a fair comparison, both methods use the same sampling configuration with $N_{\text{stratified}} = 32$ and $N_{\text{importance}} = 8$ samples per ray.

A. Localization Result

We adopt the evaluation protocol from Go-SURF [15] and report five metrics. For reconstruction, Depth L1 measures the mean absolute error between rendered and ground-truth depth. Accuracy (Acc.) and Completeness (Comp.) measure the mean distances from the reconstructed mesh to the ground truth and vice versa, respectively. Completeness Ratio reports the percentage of ground-truth surface points within a 5 cm threshold. For localization, Absolute Trajectory Error (ATE) measures the deviation between estimated and ground-truth camera poses.

Table II compares our method against the SNI-SLAM baseline on the Replica room_1 sequence. Our method incorporates uncertainty-guided keyframe selection, importance sampling, and vectorized uncertainty computation.

Our method maintains comparable Depth L1 error while significantly improving localization accuracy. The ATE Mean reduces from 0.7454 cm to 0.5368 cm, and ATE RMSE from 1.10 cm to 0.6833 cm, demonstrating that uncertainty-guided keyframe selection leads to more informative updates and better camera pose estimation. The slight drop in completeness ratio (90.44% vs. 90.73%) is attributed to the change in keyframe selection strategy. Our vectorized implementation achieves 0.68 FPS compared to 0.88 FPS for the baseline, representing a 23% overhead introduced by the uncertainty estimation pipeline.

TABLE II: Quantitative comparison on Replica room_1. Best results in **bold**.

Method	D-L1↓	Comp.(%)↑	ATE Mean↓	ATE RMSE↓	FPS↑
Baseline (w/ GT seg)	**1.21**	**90.73**	0.7454	1.10	**0.88**
Ours	**1.21**	90.44	**0.5368**	**0.6833**	0.68

As shown in Fig. 8, the trajectory estimation results demonstrate noticeable improvements in curved regions. Incorporating uncertainty-guided keyframe selection reduces the ATE RMSE from 1.10 cm to 0.6833 cm and ATE mean from 0.7454 cm to 0.5368 cm, indicating more consistent pose estimation throughout the sequence.

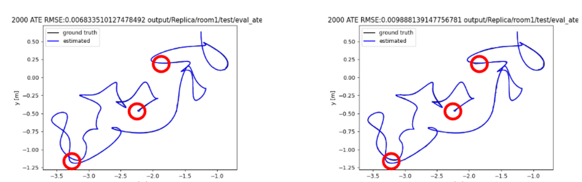

Fig. 8: Trajectory estimation comparison. Uncertainty-guided keyframe selection improves pose estimation, particularly in curved regions.

B. Mapping Results

From the reconstruction metrics and qualitative comparison shown in Fig. 9, we observe no significant difference between the two methods. This is primarily because the number of samples is fixed, which limits the achievable reconstruction quality. We further experiment with varying the number of samples along each ray and find that increasing either the stratified samples or the importance samples improves reconstruction quality, but at the cost of additional computation time. For a fair performance comparison, we fix the number of stratified and importance samples to 32 and 8, respectively, and focus on evaluating other performance improvements.

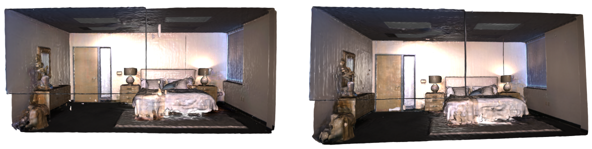

Fig. 9: Comparison of reconstruction results between our method and the baseline. No significant visual difference is observed.

VI. Discussion

From the convergence results, the model exhibits clear signs of underfitting. Increasing the number of samples along the ray consistently improves reconstruction quality, suggesting that the NeRF representation benefits from denser geometric supervision from RGB-D input. This highlights a fundamental tension in online neural SLAM: each frame is observed only once during the mapping process, which limits the model's ability to fully represent fine-grained scene details. However, iterating too many times per frame would undermine the real-time nature of the system, effectively transforming the problem from online NeRF-SLAM into offline NeRF reconstruction.

Our results demonstrate that incorporating rendering uncertainty into keyframe selection accelerates convergence and improves both reconstruction and localization metrics, at the cost of additional computation even with our vectorized batch implementation. This trade-off reflects a broader challenge in neural SLAM: balancing representational capacity with computational efficiency.

Nevertheless, our work validates the importance of rendering uncertainty as a signal for guiding scene learning in SLAM. The improvements in localization metrics confirm that uncertainty-aware frame selection leads to more informative gradient updates, enabling the network to prioritize geometrically ambiguous regions. This opens the door to more powerful architectures that leverage uncertainty in a principled manner while maintaining real-time efficiency.

VII. Conclusion

In this work, we presented an uncertainty-guided keyframe selection framework for neural implicit SLAM, building upon SNI-SLAM. By computing per-frame rendering uncertainty via Shannon entropy of volume rendering weights, our method identifies frames that contribute the most information to the scene representation and prioritizes them during mapping optimization. We further introduced a vectorized batch uncertainty computation that reduces the per-frame overhead by processing all keyframes in a single decoder forward pass.

Experiments on the Replica dataset demonstrate that our approach consistently improves localization accuracy while maintaining comparable reconstruction quality, confirming that uncertainty-aware keyframe selection leads to more effective scene updates and better camera pose estimation. Despite the computational overhead introduced by uncertainty estimation, our vectorized implementation effectively mitigates this cost.

Future work may explore more efficient uncertainty proxies, adaptive sampling strategies driven by spatial uncertainty maps, and extensions to dynamic scenes or large-scale environments where principled frame selection becomes even more critical.

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